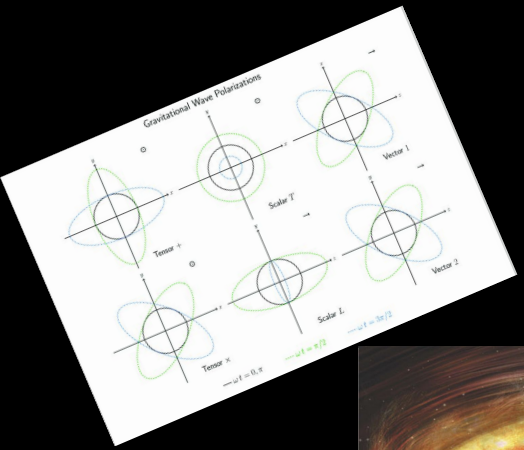
Testing Massive Gravity with Pulsar Timing Arrays

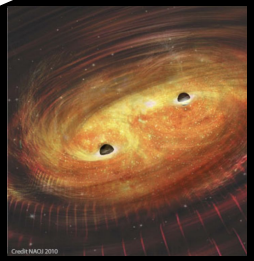
2610.31912

•••

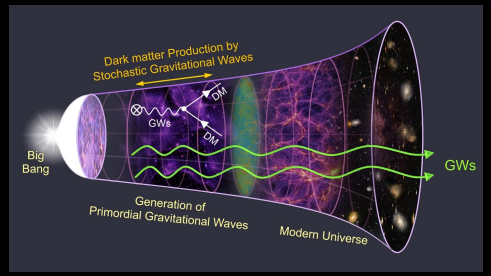
Chris Choi

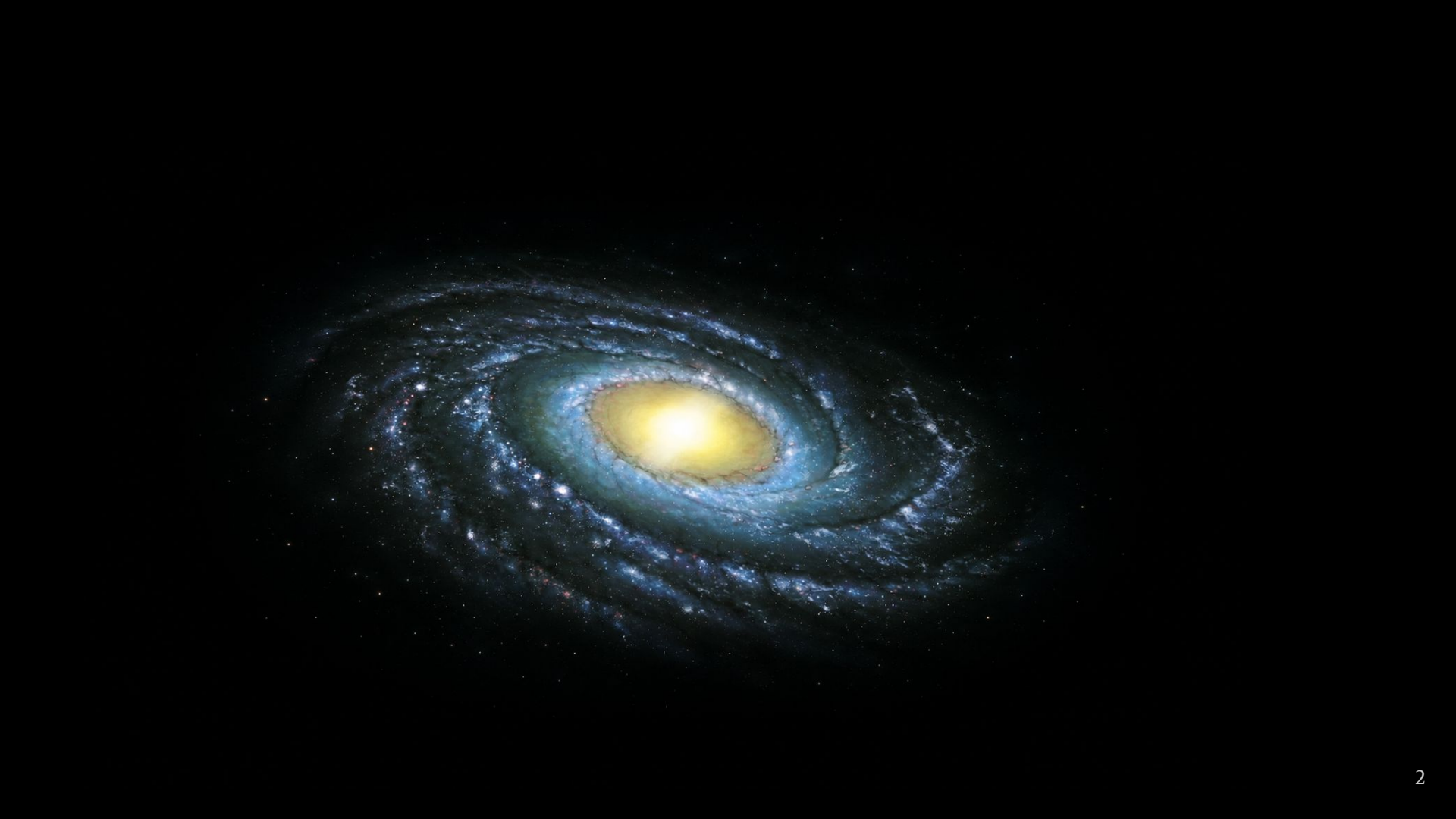


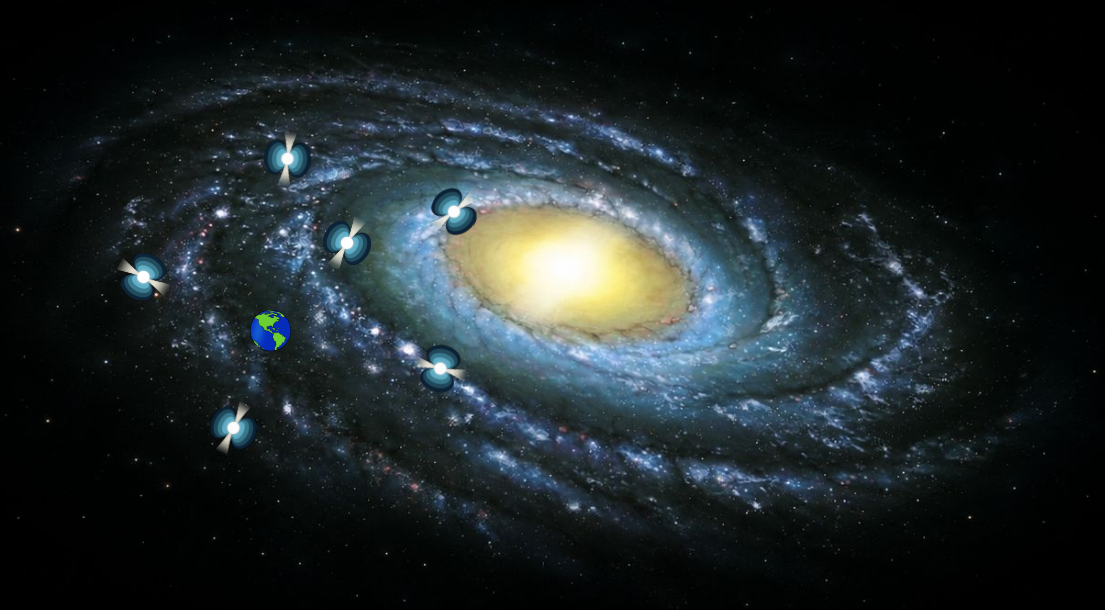




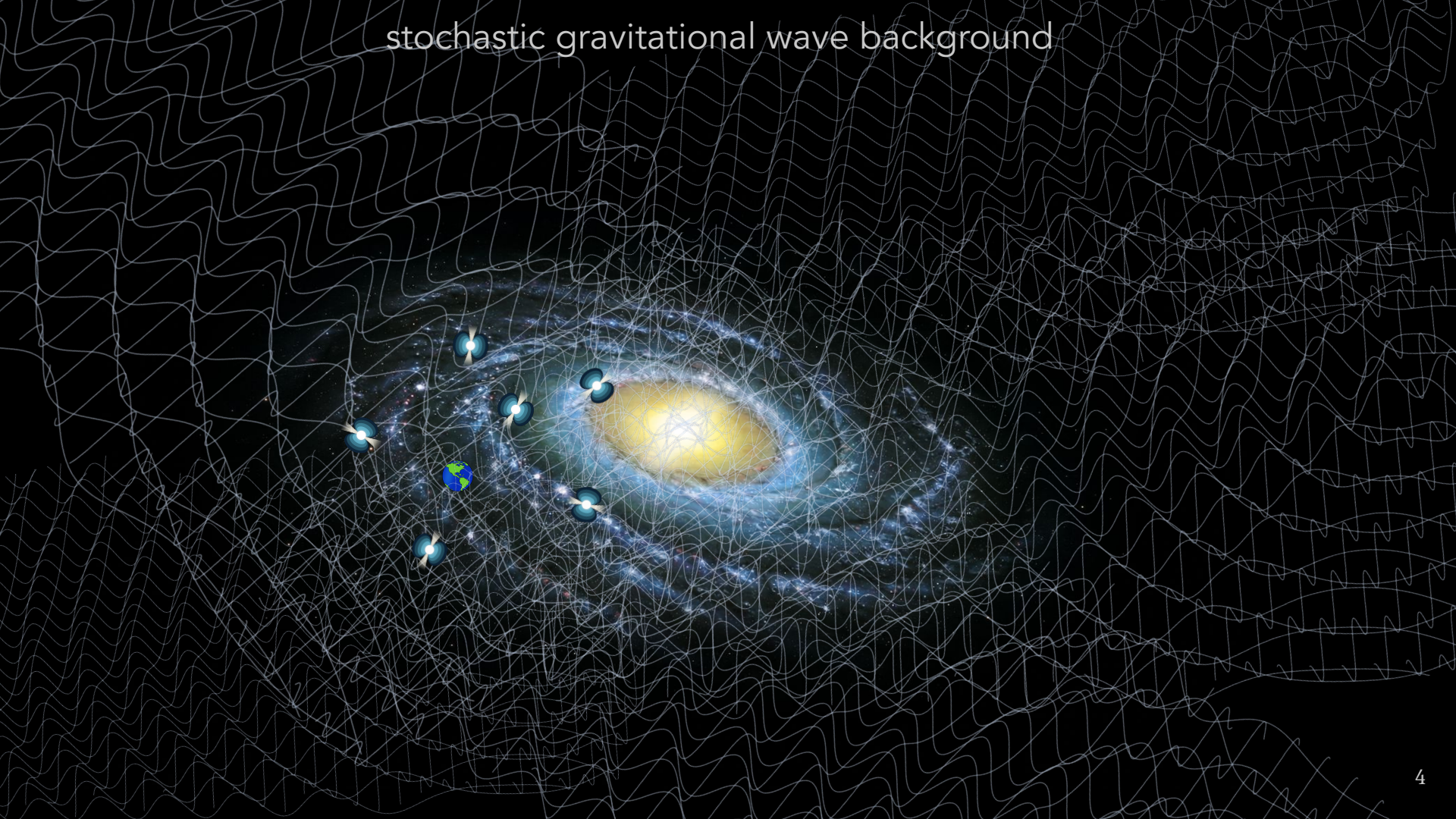
KAAI
Workshop 2





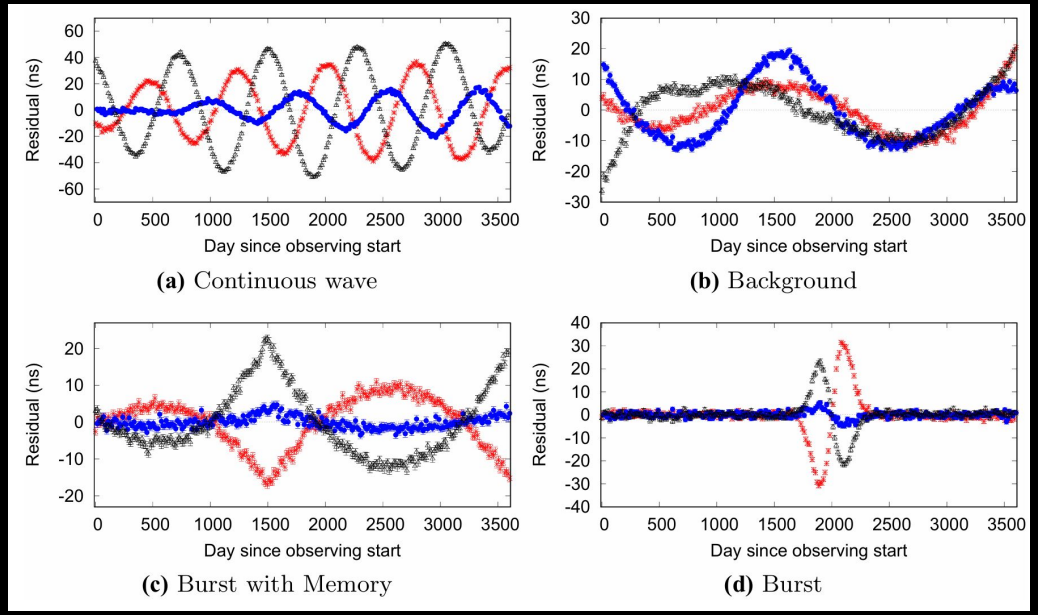


stochastic gravitational wave background



pulsars have intrinsic noise

Burke-Spolaor et al., 2018 [1811.08826]



Each panel shows pulsar timing residuals for three pulsars (black triangles, red stars, blue circles)

we cross-correlate timing residuals
to find the GW background

$$\langle \tilde{z}_a(f) \tilde{z}_b^*(f') \rangle \propto \delta(f - f') \frac{\Omega_{\rm GW}(f)}{f^3} \Gamma_{ab}(\xi_{ab})$$

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energy density

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energy density

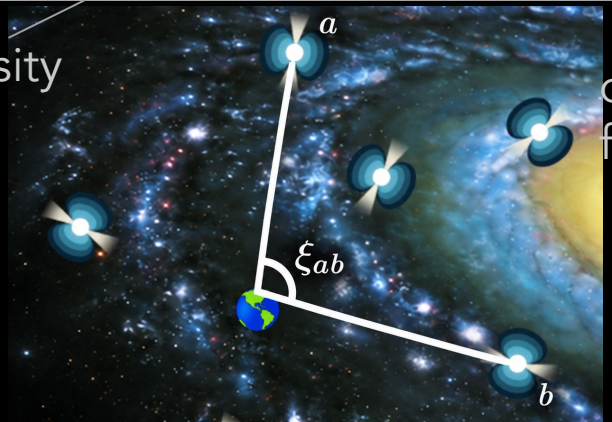
overlap reduction function (ORF)

we cross-correlate timing residuals to find the GW background

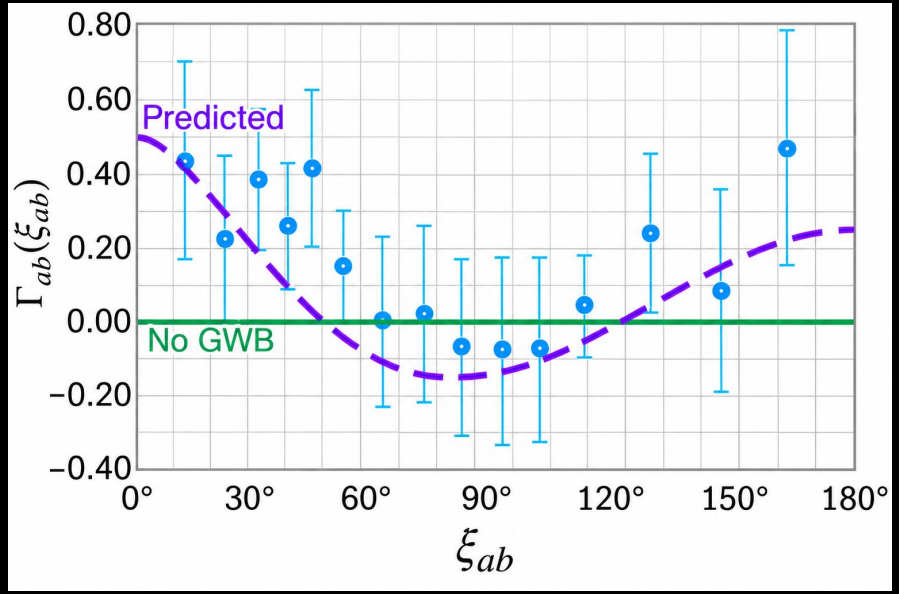
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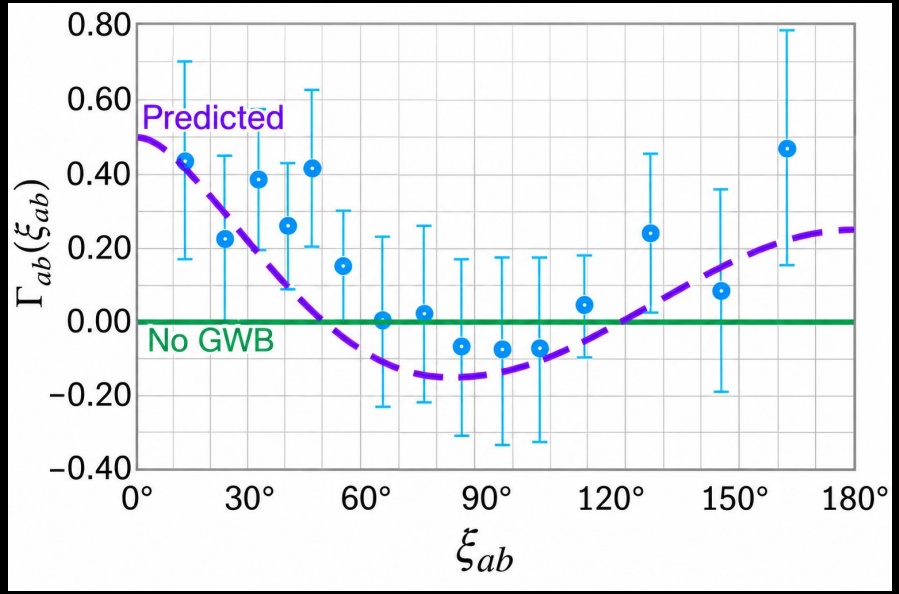


GR has 2 tensor polarizations, predicts Hellings-Downs correlation



Agazie et al., 2023 [2306.16213] Fig 1c:

GR has 2 tensor polarizations, predicts Hellings-Downs correlation



$$\widetilde{\Gamma}_{ab}^{\mathrm{HD}}(\xi_{ab}) = \frac{3}{2}x\ln(x) - \frac{1}{4}x + \frac{1}{2} + \frac{1}{2}\delta_{ab}$$

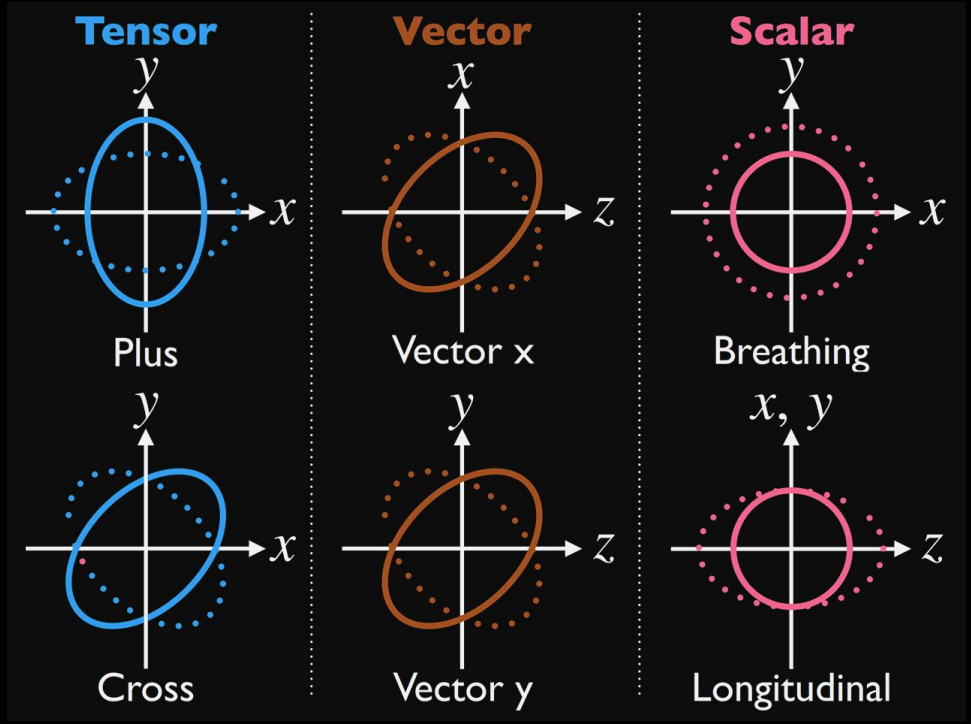
$$x = \frac{1-\cos\xi_{ab}}{2}.$$

Hellings, Downs 1983

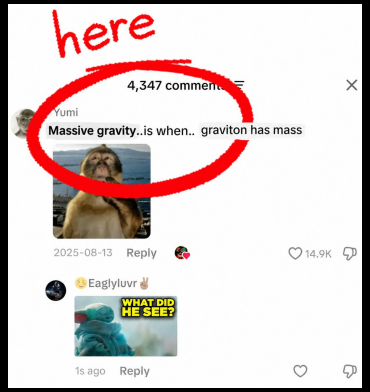
Agazie et al., 2023 [2306.16213] Fig 1c:

generically, we have up to 6 GW polarizations

Streibart et al., 2025
[2404.07782]

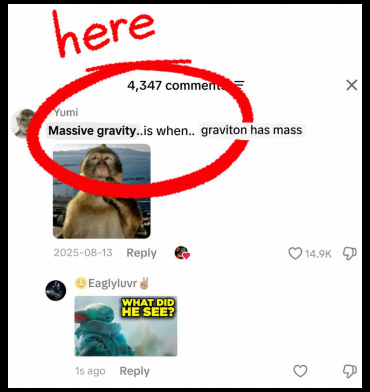


massive gravity is when graviton has mass



5 polarizations (2 scalar -> 1 degree of freedom)

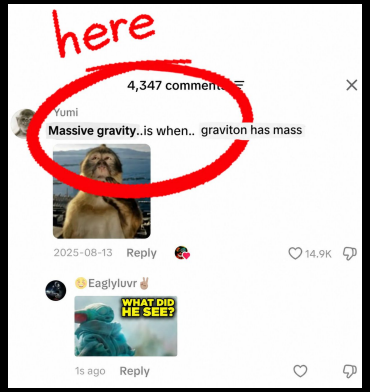
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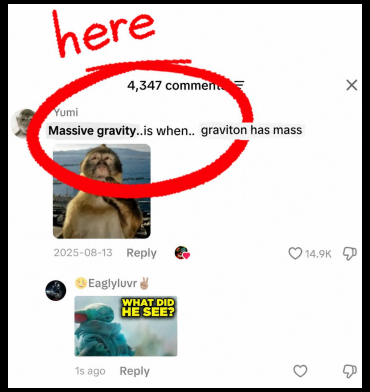


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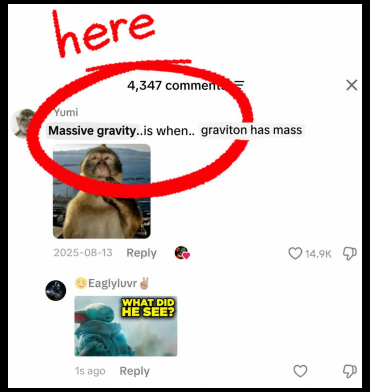
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modified behavior of energy spectrum and ORFs

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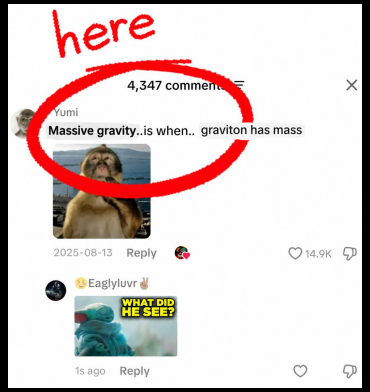
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is gravity truly long ranged?

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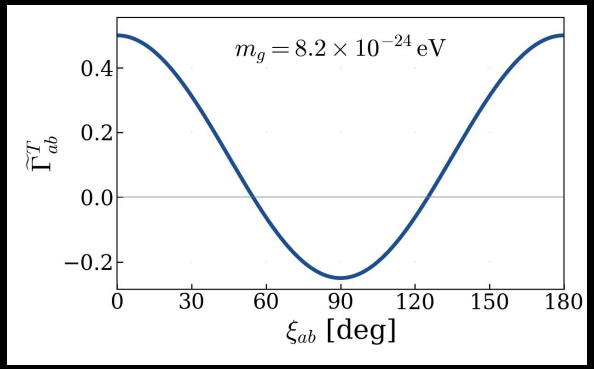
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is gravity truly long ranged?

problems emerge when taking massless limit

we should recover GR, but it's not so straightforward

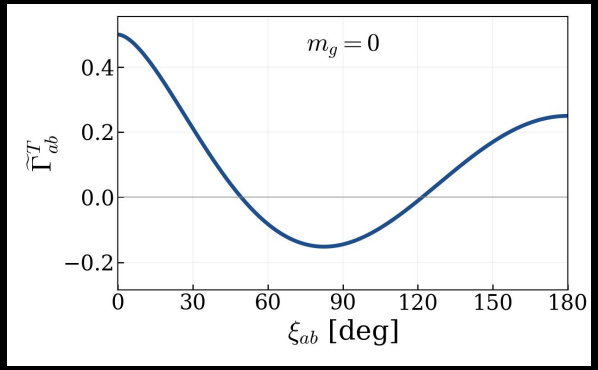
tensor



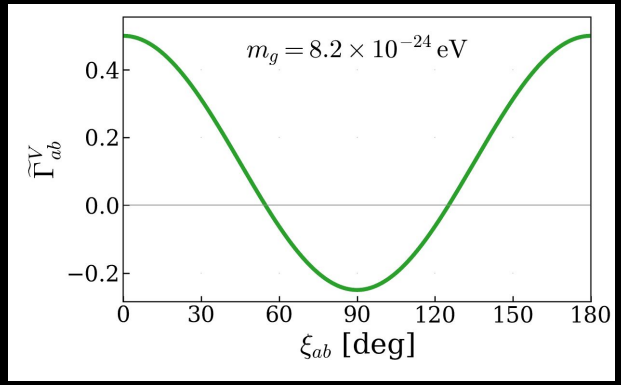


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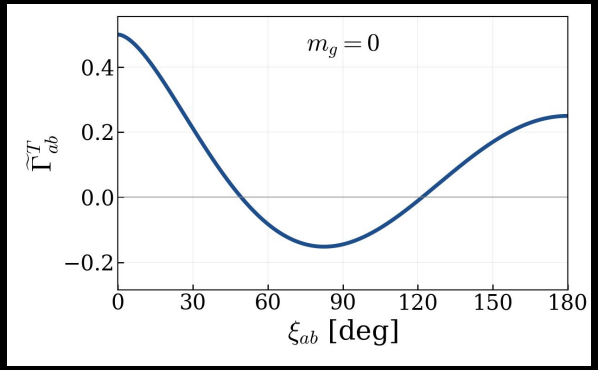
vector



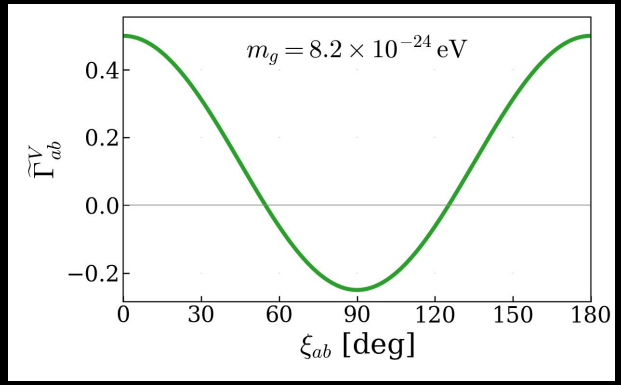


we should recover GR, but it's not so straightforward

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vector

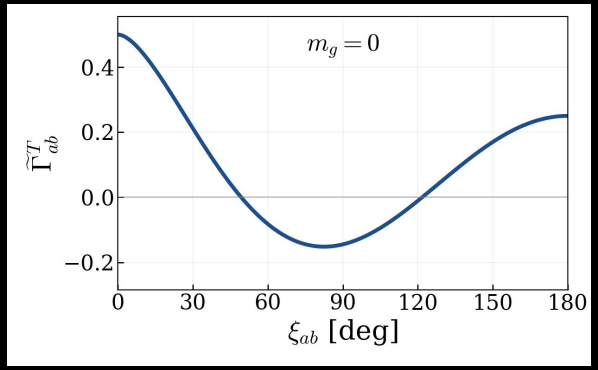


$$\widetilde{\Gamma}^V_{ab} = \beta_T \left(\frac{m_g}{2\pi f}\right)^2 \mathrm{Re} \int d\Omega\, \mathcal{K}^V_{ab}\left(\hat{\mathbf{\Omega}};\, m_g, f, L_a, L_b, \xi_{ab}\right)$$

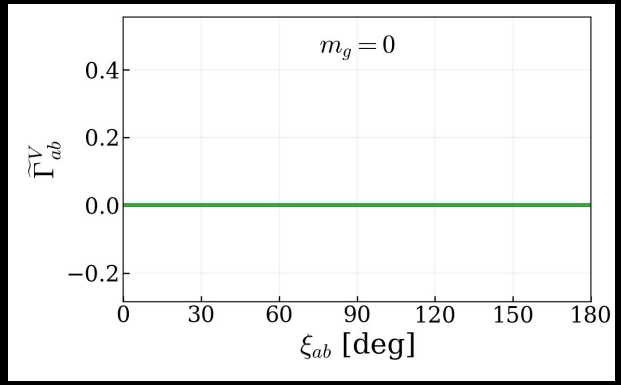


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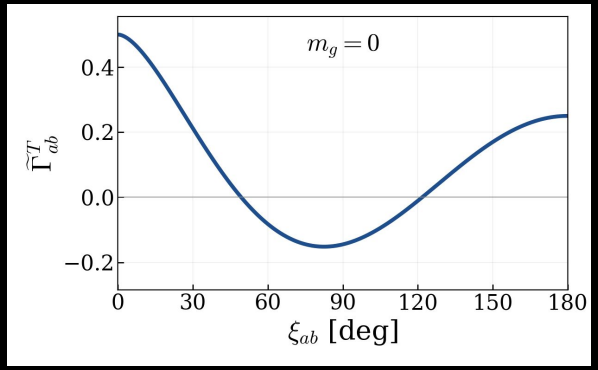
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goes to 0

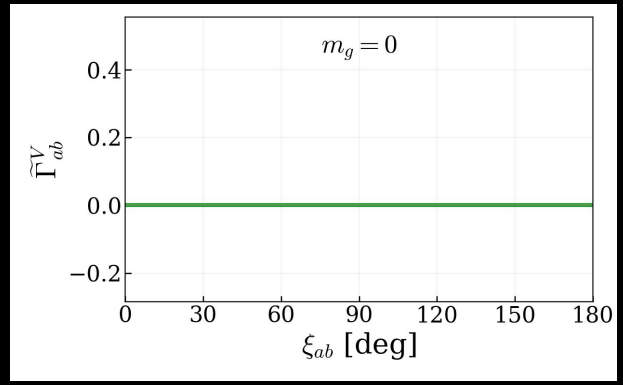


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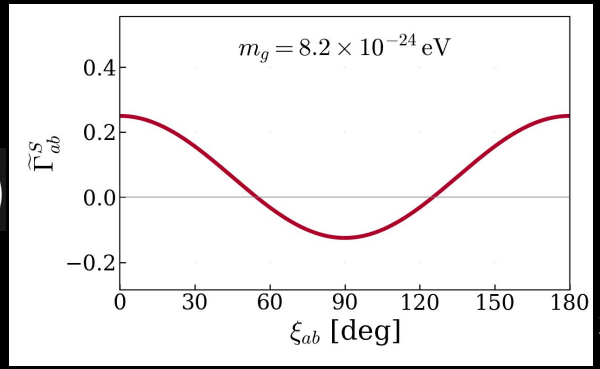


vector





scalar



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scalar mode remains in massless limit due to vDVZ discontinuity

van Dam, Veltman, & Zakharov 1970

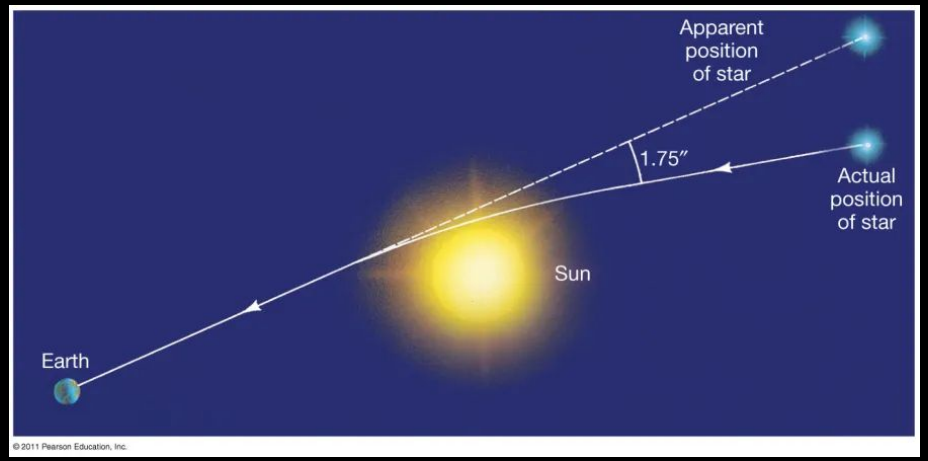
in linear massive gravity, scalar mode couples to matter, even in the massless limit!

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general relativity

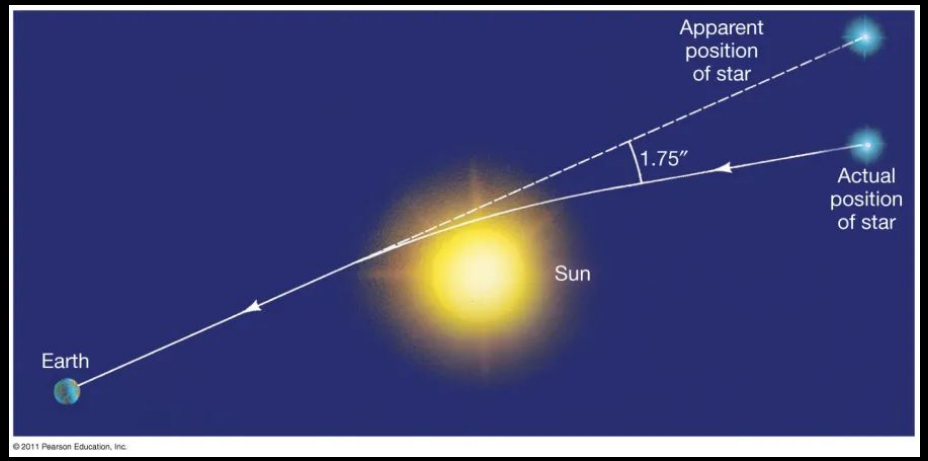


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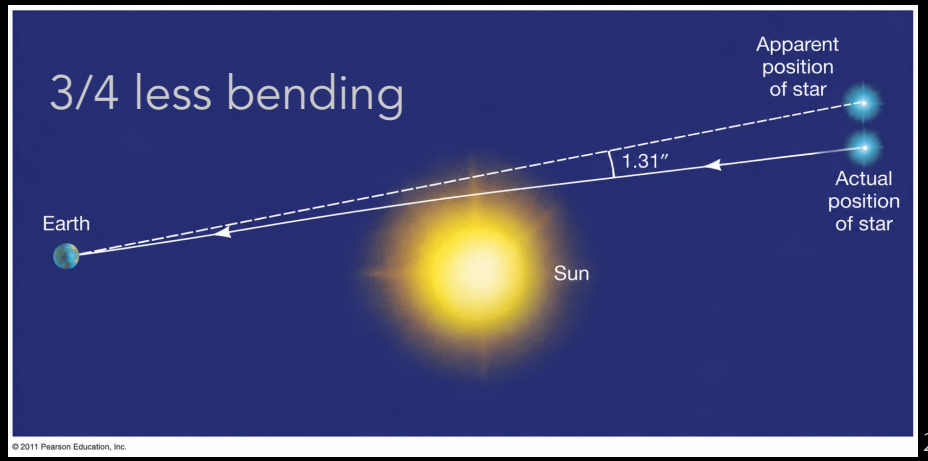
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vDVZ case



nonlinearities help recover GR

need nonlinearities to decouple scalar mode

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need nonlinearities to decouple scalar mode

example: cubic Galileon model for static, symmetric source

$$\underbrace{\frac{\pi'}{r}}_{\text{Linear}} + \underbrace{\frac{1}{\Lambda_3^3}\left(\frac{\pi'}{r}\right)^2}_{\text{Nonlinear}} = \frac{M}{4\pi M_{\rm Pl} r^3}$$

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new scale: Vainshtein radius

$$r_V \sim \left(\frac{GM}{m_g^2}\right)^{1/3}$$

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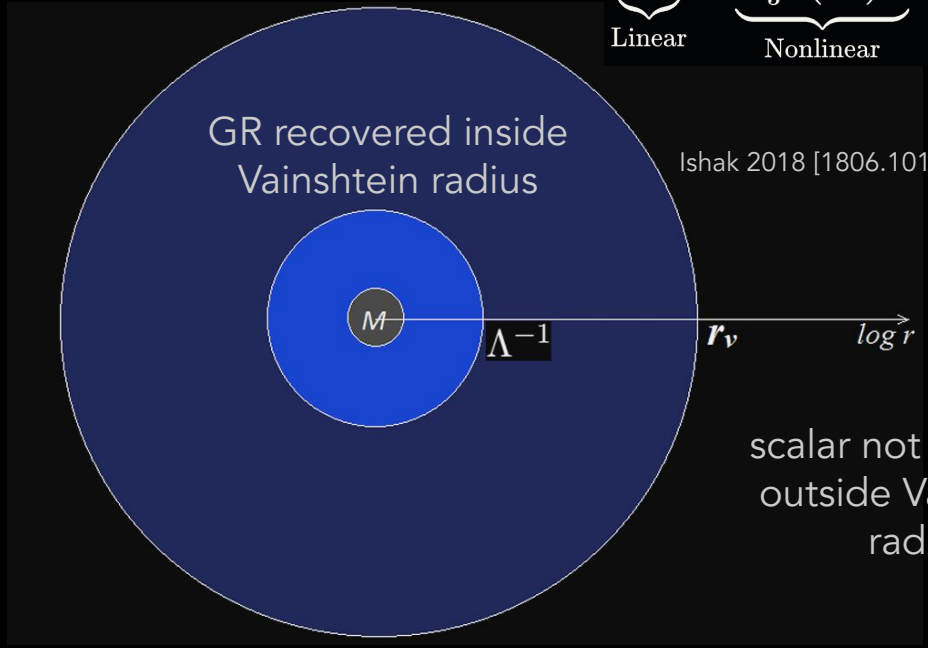
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Ishak 2018 [1806.10122]

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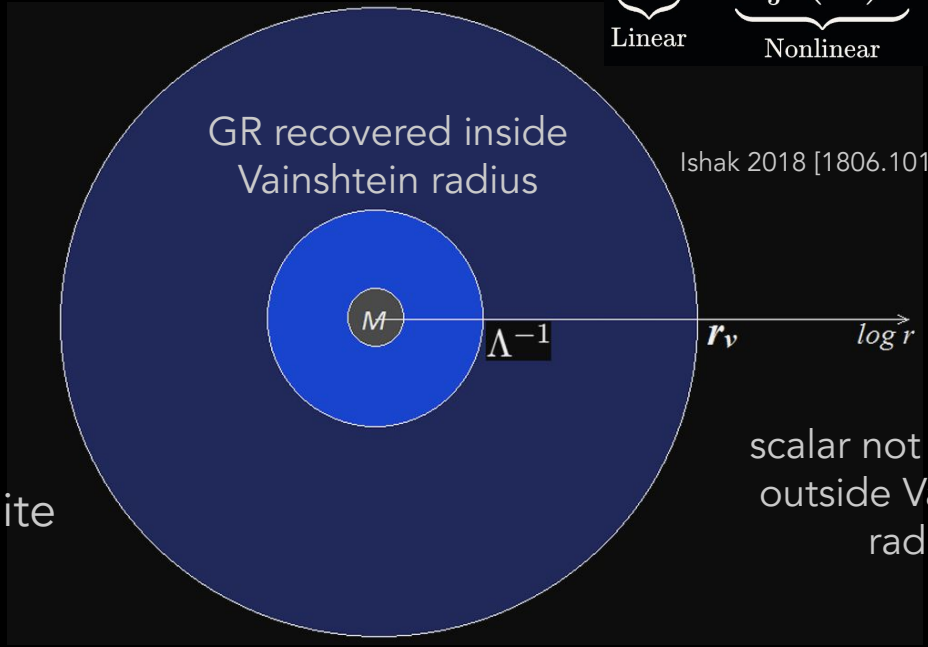
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Vainshtein radius becomes infinite when graviton mass goes to 0



Ishak 2018 [1806.10122]

scalar not screened outside Vainshtein radius

suppression of scalar ORF comes from energy density

effective ORF

$$\widetilde{\Gamma}_{ab}^{\mathrm{eff}} = \beta_T \left[\Gamma_{ab}^{T} + \frac{\Omega_V}{\Omega_T} \Gamma_{ab}^{V} + 2 \frac{\Omega_S}{\Omega_T} \Gamma_{ab}^{S} \right]$$

suppression of scalar ORF comes from energy density

effective ORF

$$\widetilde{\Gamma}^{\rm eff}_{ab} = \beta_T \left[\Gamma^T_{ab} + \frac{\Omega_V}{\Omega_T} \Gamma^V_{ab} + 2 \frac{\Omega_S}{\Omega_T} \Gamma^S_{ab} \right]$$

tensor normalization

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this must go to 0 in massless limit

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phenomenological suppression

propose power law prescription

$$\frac{\Omega_S}{\Omega_T} = A_S \left(\frac{m_g}{\omega}\right)^{n_S}$$

phenomenological suppression

propose power law prescription

scalar amplitude

suppression index

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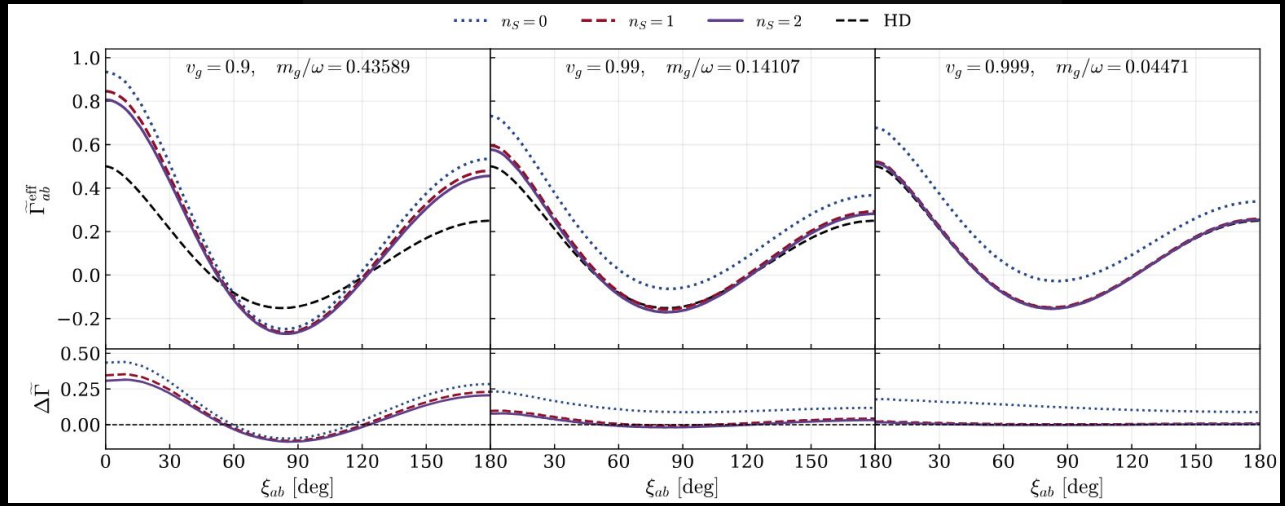
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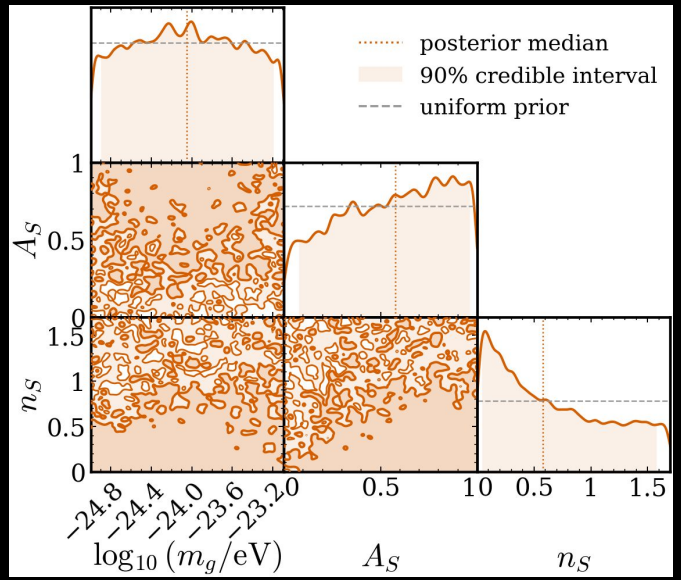
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Choi et al., 2026 [2610.31912]



these parameters are poorly constrained by NANOGrav

Param.	Description	Prior
m_g/eV	Graviton mass [eV]	log-U$[10^{-25},10^{-23.09}]$
A_S	Scalar-tensor relative amp.	U$[0,1]$
n_S	Scalar suppression index	U$[0,1.7]$

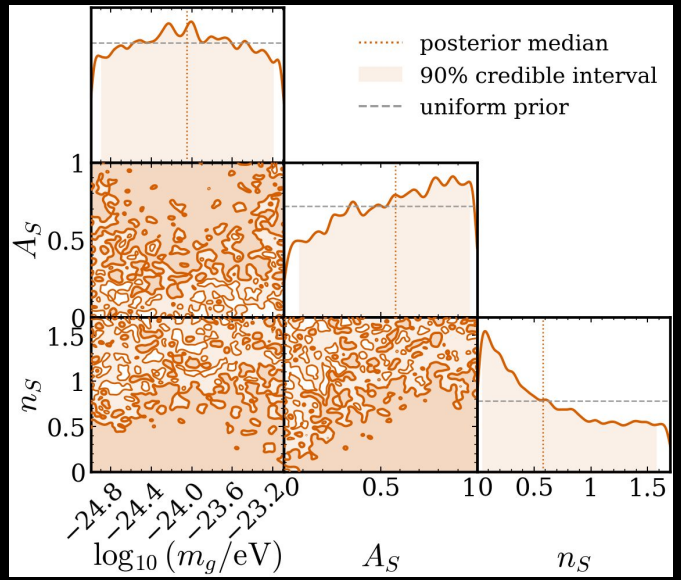


NANOGrav

work in progress w/ Marcus Bosca
Tina Kahniashvili, Barnabas Poczos

these parameters are poorly constrained by NANOGrav

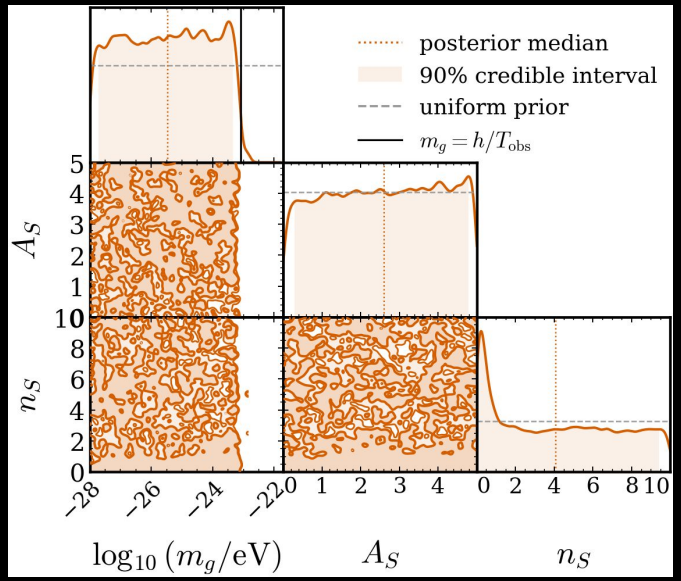
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NANOGrav

work in progress w/ Marcus Bosca
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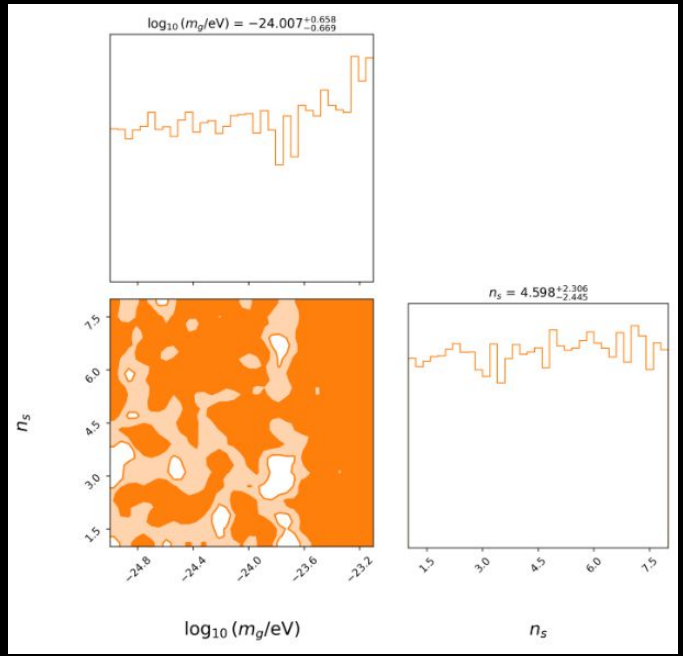
Param.	Description	Prior
m_g/eV	Graviton mass [eV]	log-U$[10^{-28},2\times10^{-22}]$
A_S	Scalar-tensor relative amp.	U$[0,5]$
n_S	Scalar suppression index	U$[0,10]$



also poorly constrained by EPTA

ongoing work w/ Marcus Bosca, Alberto Roper Pol, Tina Kahniashvili

Param.	Description	Prior
m_g/eV	Graviton mass [eV]	log-U$[10^{-25},10^{-23.1}]$
n_S	Scalar suppression index	U$[1,8]$



also poorly constrained by EPTA



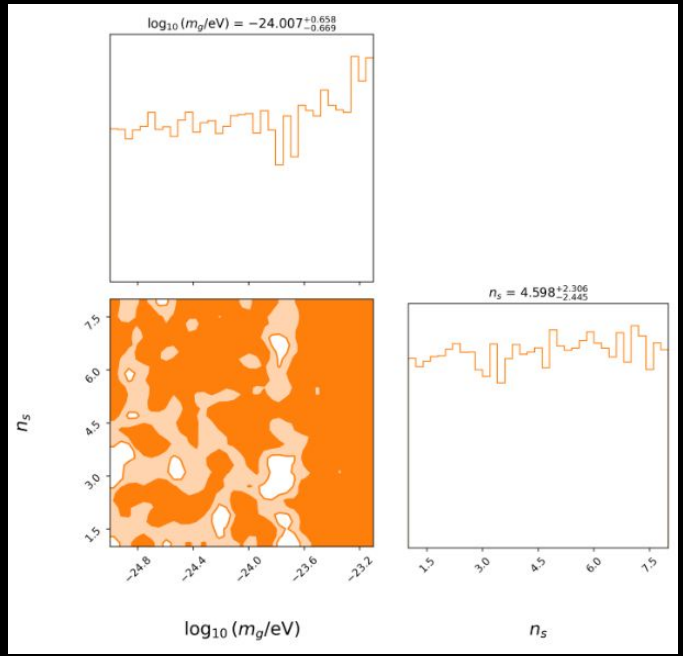
ongoing work w/ Marcus Bosca, Alberto Roper Pol, Tina Kahniashvili

two possibilities

prior may be misspecified

data isn't sensitive to massive gravity

Param.	Description	Prior
m_g/eV	Graviton mass [eV]	log-U$[10^{-25},10^{-23.1}]$
n_S	Scalar suppression index	U$[1,8]$



what do we need to assume in order to explicitly infer suppression parameters

assumptions

assume the nonlinear theory of massive gravity

cubic galileon

what do we need to assume in order to explicitly infer suppression parameters

assumptions

assume the nonlinear theory of massive gravity

cubic galileon

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supermassive black hole binaries

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assume that GWs are linear perturbations to nonlinear background

suppression from binary black hole in Vainshtein screening

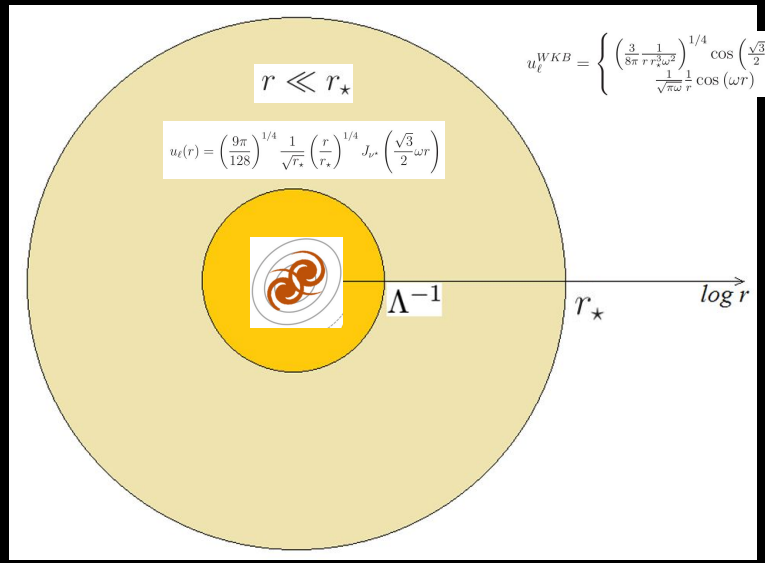
Vainshtein Mechanism in Binary Pulsars

Claudia de Rham[a], **Andrew J. Tolley**[a], **Daniel H. Wesley**

a Department of Physics, Case Western Reserve University, 10900 Euclid Ave, Cleveland, OH 44106, USA

galileon field, keep up to cubic

$$S = \int \mathrm{d}^4x\sqrt{-g}\left[\frac{M_{\rm Pl}^2}{2}R - \frac{3}{4}(\partial\pi)^2\left(1+\frac{1}{3\Lambda^3}(\Box\pi)\right)\right]$$



$$u_\ell^{WKB} = \begin{cases} \left(\frac{3}{8\pi}\frac{1}{r r_\star^2\omega^2}\right)^{1/4}\cos\left(\frac{\sqrt{3}}{2}\omega r\right) & \text{for } \omega^{-1} \ll r \ll r_\star \\ \frac{1}{\sqrt{\pi\omega}}\frac{1}{r}\cos(\omega r) & \text{for } r \gg r_\star \end{cases}$$

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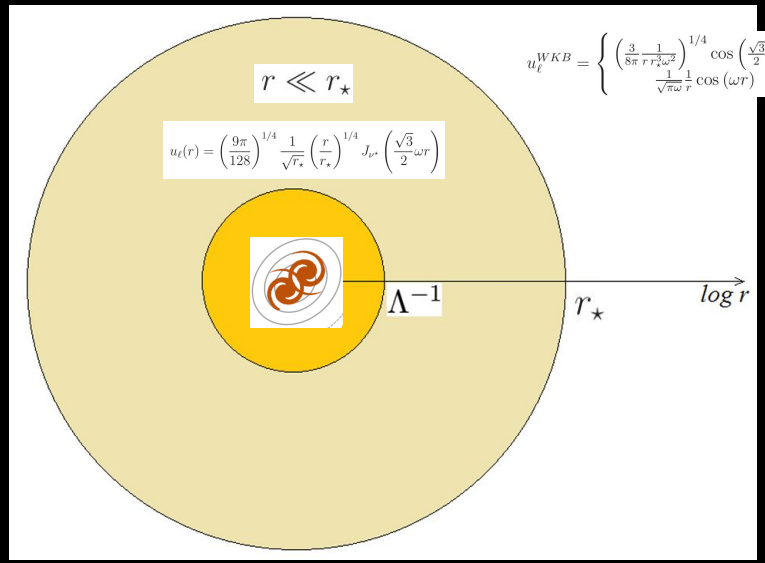
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GR quadrupole radiation

$$P^{\rm GR}_{\rm quadrupole} = \frac{4}{5\pi M_{\rm Pl}^2}\frac{(1+\frac{73}{24}e^2+\frac{37}{96}e^4)}{(1-e^2)^{7/2}}(\Omega_P\bar{r})^4\frac{M_1^2M_2^2}{(M_1+M_2)^2}\Omega_P^2$$

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$$P^{\rm Galileon}_{\rm quadrupole} = \frac{5\lambda^2}{32}\frac{(\Omega_P\bar{r})^3}{(\Omega_P r_\star)^{3/2}}\frac{M_Q^2}{M_{\rm Pl}^2}\Omega_P^2\sum_{n=0}^{\infty}\left|I_n^Q(e)\right|^2$$



suppression from binary black hole in Vainshtein screening

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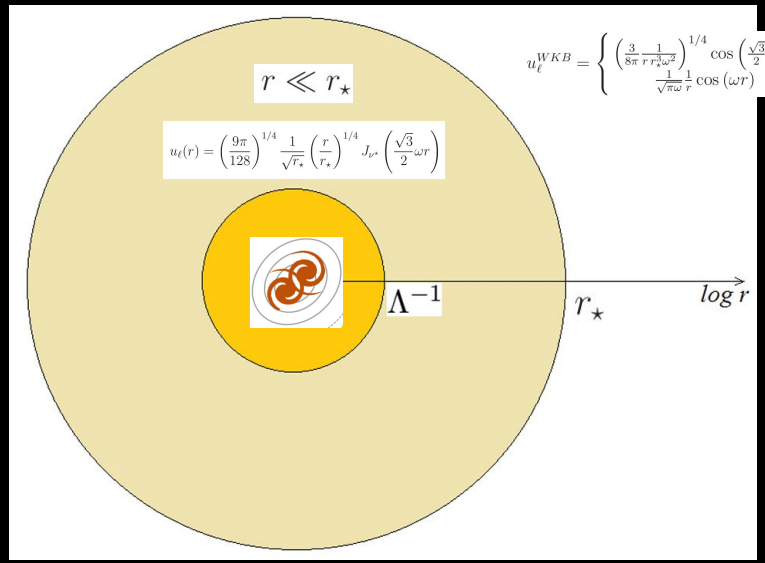
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population average

A practical theorem on gravitational wave backgrounds

E.S. Phinney*
Theoretical Astrophysics, 130-33 Caltech, Pasadena, CA 91125, USA

2001 July 31

$$\Omega_{\rm gw}(f) = \frac{1}{\rho_{\rm crit}}\int_0^\infty N(z) f\left(\frac{P_{\rm gw}}{\dot{f}_r}\right)\Big|_{f_r=f(1+z)} dz$$

assume *N(z)* based on merger rates from Chen, 2018 [1810.04184]

$$\frac{\Omega_S(f)}{\Omega_T(f)} \simeq A_S^{\rm SW}\frac{m_g}{\omega}$$

$n_S = 1$

$$A_S^{\rm SW} \simeq 7.08\times10^2\left(\frac{f}{1\,\mathrm{yr}^{-1}}\right)^{-5/6}$$

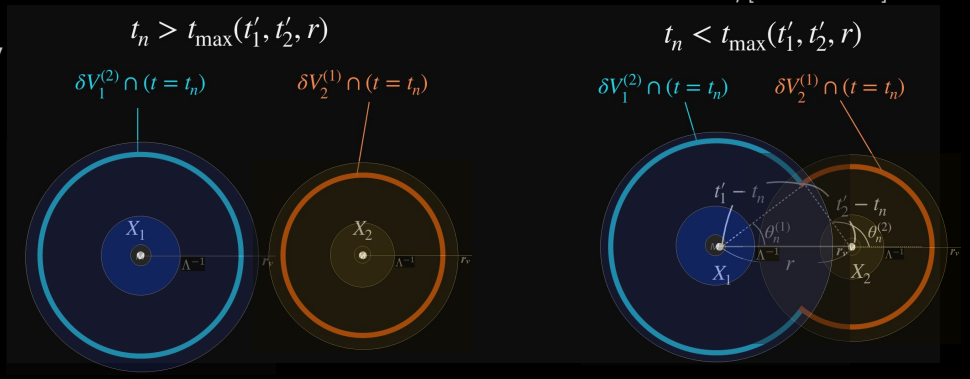
frequency-dependence in the amplitude?!?!?

connections to this workshop

can we compute the scalar energy density in early universe GW source mechanisms?

- 1st order phase transitions
- turbulent sources

Inomata 2024, [2412.17912]

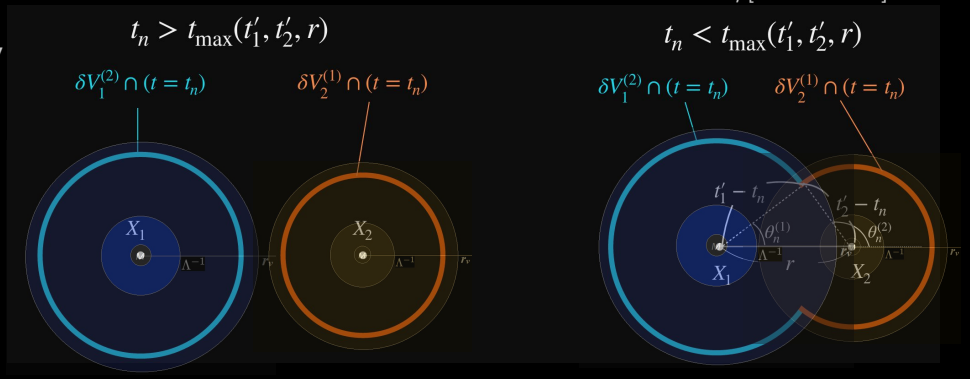


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↓

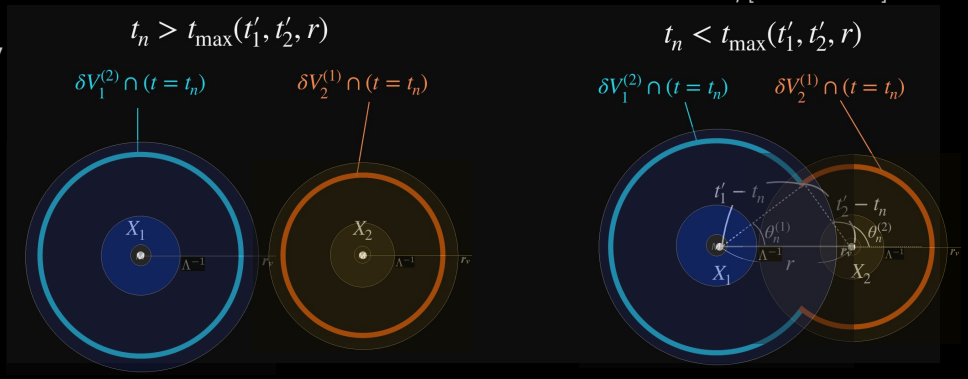
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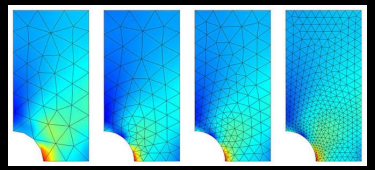
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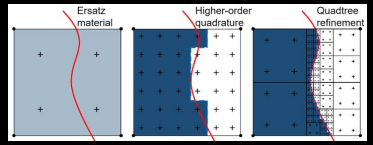
↓

using a fixed mass, can we generate realizations of the PTA timing data and infer the graviton mass?

sampling continuous mass is *very* expensive, can we use adaptive mesh or adaptive integrators using ML?

multiphysics-cyclopedia





Swierstra et al, 2020

conclusion

main takeaway: I am interested in connecting nonlinear dynamics of massive gravity to PTA data

Thank you for listening!

